



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Advanced)

Full Syllabus

31-07-2021

JEE(Main + Advanced) : Leader & Enthusiast Course (PHASE : SCORE-3.0)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I(i)	Q.	1	2	3	4	5	6		
	A.	A,B	B,C	B,C	B,D	A,D	B,D		
SECTION-I(ii)	Q.	7	8	9	10				
	A.	C	A	B	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	15.00	15.00	10.00	37.5	129.6	45.00	18.00	4.00

PART-2 : CHEMISTRY

SECTION-I(i)	Q.	1	2	3	4	5	6		
	A.	A,C,D	A,B,C,D	A,C,D	B,C	A,B,C	A,B,D		
SECTION-I(ii)	Q.	7	8	9	10				
	A.	B	D	D	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	3.00	5.00	37.50	4.00	6.00	7.00	10.00	10.00

PART-3 : MATHEMATICS

SECTION-I(i)	Q.	1	2	3	4	5	6		
	A.	C,D	B,C,D	A,B,C	A,C	A,B,C	A,D		
SECTION-I(ii)	Q.	7	8	9	10				
	A.	D	C	A	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	1.00	2.50	4.00	0.25	156.00	3.00	1.00	190.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I(i)

1. Ans (A,B)

At $t = \frac{u}{a}$ relative velocity is zero. Elongation of spring at that instant

$$x = ut - \frac{1}{2}at^2 = \frac{u^2}{2a}$$

$$U_{sp} = \frac{1}{2}kx^2 = \frac{ku^4}{8a^2}$$

Also, $W_A + W_B = U_{sp}$ and $W_B = \int_0^{W_A} kx \, u \, dt$

2. Ans (B,C)

Conserving angular momentum

$$mVr = mV_i r_i \sin 30^\circ$$

$$\Rightarrow Vr = 1$$

Using mechanical energy conservation

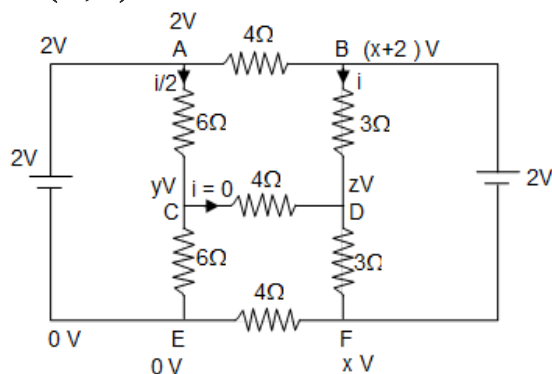
$$\frac{1}{2}mV^2 + \frac{1}{2}Kr^2 = \frac{1}{2}mV_i^2 + \frac{1}{2}Kr_i^2$$

$$\therefore \frac{1}{r^2} + r^2 = 4 + 1$$

$$\therefore \left(\frac{1}{r} + r\right) = \pm\sqrt{5}$$

$$\therefore r_{\max} = \frac{\sqrt{5} + \sqrt{3}}{2}, r_{\min} = \frac{\sqrt{5} - \sqrt{3}}{2}$$

3. Ans (B,C)



Apply junction rule at B, C and D

$$\frac{2-x-2}{4} + \frac{z-x-2}{3} = \frac{x-z}{3} + \frac{x-0}{4}$$

$$-\frac{x}{4} + \frac{z-x}{3} - \frac{2}{3} = \frac{x-z}{3} + \frac{x}{4}$$

$$4x - 4z + 3x = -4$$

$$7x - 4z = -4$$

$$\left(\frac{2-y}{6} = \frac{y-z}{4} + \frac{y-0}{6} \right)$$

$$4 - 2y = 3y - 3z + 2y$$

$$7y - 3z = 4$$

$$\left(\frac{y-z}{4} + \frac{x+2-z}{3} = \frac{z-x}{3} \right)$$

$$3y - 3z + 4x + 8 - 4z = 4z - 4x$$

$$8x + 3y - 11z = -8$$

Solving Above equation we will get

$$x = 0, y = 1, z = 1$$

Due to symmetrical circuit, potential across 4Ω is zero and current flow will be as shown in the figure.

$$\text{Hence current through } 6\Omega \text{ is } = \frac{2}{12} = \frac{1}{6} \text{ A}$$

$$\text{and current through } 3\Omega \text{ is } = \frac{2}{6} = \frac{1}{3} \text{ A}$$

4. Ans (B,D)

$$\text{Flux through frame} = \frac{\mu_0 i}{2\pi} \int_2^4 \frac{4dx}{x}$$

$$\therefore \phi = \left(\frac{2\mu_0 \ell n2}{\pi} \right) i$$

$$\therefore \epsilon = \left| \frac{d\phi}{dt} \right| = \left(\frac{2\mu_0 \ell n2}{\pi} \right) \frac{di}{dt}$$

$$= \frac{2\mu_0 i_0 \omega \ell n2}{\pi} \cos\left(\omega t + \frac{\pi}{6}\right)$$

$$\therefore Q_{\max} = \epsilon_{\max} C = \left(\frac{2\mu_0 i_0 \omega \ell n2}{\pi} \right) C$$

$$\text{and } i_{\text{frame}} = \left| \frac{dq}{dt} \right| = \left(\frac{2\mu_0 i_0 \omega^2 \ell n2}{\pi} \right) C \sin\left(\omega t + \frac{\pi}{6}\right)$$

6. Ans (B,D)

Smaller the least count greater the precision.

$$\text{Mean absolute error } |\Delta x| = \frac{\sum |x - x_i|}{N}$$

SECTION-I(ii)

8. Ans (A)

The unknown products are 6 beta (${}_{-1}^0\text{e}$)

particles. In atomic mass units :

$$Q = (m_U - 8m_{\text{He}} - m_{\text{Pb}})C^2 = (238.050783 - 8 \times 4.002603 - 205.974449)u \times (931.5 \text{ MeV/u}) = 51.7 \text{ MeV}$$

If activity of 100 kg sample is A, then power released will be $P = AQ \lambda =$

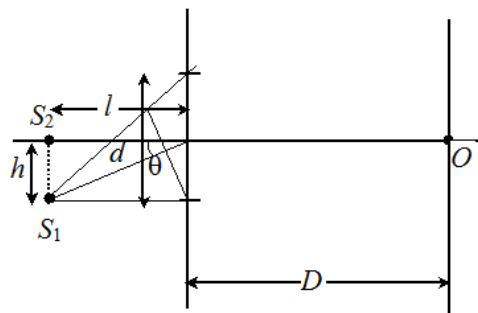
$$\frac{\ell n2}{T_{1/2}} = \frac{0.693}{4.47 \times 10^9} = 1.55 \times 10^{-10} \text{ yr}^{-1}$$

$$\text{Population } N = \frac{100 \times 10^3}{238} \times 6.02 \times 10^{23}$$

$$\therefore A = \lambda N = 3.92 \times 10^{16} \text{ decays/yr}$$

$$\therefore P = AQ = 0.01 \text{ Watt}$$

10. Ans (C)



Here

$$(a) \Delta x = d \sin \theta = d \tan \theta = d \times \frac{h}{l}$$

For green light to be missing

$$\Delta x = \frac{n\lambda_g}{2} \Rightarrow d \frac{h}{l} = \frac{n\lambda_g}{2} (n = \text{odd.})$$

for minimum h, n should be equal to 1
or

$$h_{\min} = \frac{\lambda_g l}{2d} = \frac{5 \times 10^{-7} \times 0.5}{2 \times 10^{-3}} = 1.25 \times 10^{-4} \text{ m}$$

Here fringe width

$$\beta = \frac{\lambda_g D}{d} = \frac{5 \times 10^{-7} \times 1}{10^{-3}} = 5 \times 10^{-4} \text{ m}$$

(b) Let CM at height Y_C from O

$$\Delta x = d \frac{h}{l} - d \frac{Y_C}{D} = 0$$

$$Y_C = 2h = 4 h_{\min}$$

$$= 5 \times 10^{-4} \text{ m}$$

$$\therefore \beta = 5 \times 10^{-4} \text{ m}$$

So first maxima at O

SECTION-II

1. Ans (15.00)

Potential at a distance y from the center

$$V = - \int_R^{4R} \frac{(2\pi x dx) \sigma G}{\sqrt{x^2 + y^2}}$$

$$= -2\pi\sigma G \left[\sqrt{16R^2 + y^2} - \sqrt{R^2 + y^2} \right]$$

$$\therefore v_i = -2\pi\sigma G (4\sqrt{5}R - \sqrt{65}R)$$

$$v_f = -2\pi\sigma G (4R - R)$$

$$\Delta u = m \Delta v = -2\pi\sigma G m [3 - \sqrt{5}(4 - \sqrt{13})]R$$

$$\Delta k = \frac{1}{2} m (v_f^2 - v_i^2); \Delta k + \Delta u = 0$$

Solving: N = 13

2. Ans (15.00)

$$\frac{r_d}{f} = \frac{R_s}{D} \quad (r_d \rightarrow \text{radius of image/disc})$$

$R_s \rightarrow$ radius of sun

$D \rightarrow$ Distance b/w earth & sun)

$$\frac{\sigma 4\pi R_s^2 T_s^4}{4\pi D^2} \times \frac{\pi d_1^2}{4} = \sigma 2\pi r_d^2 T_d^4 = \sigma 2\pi \frac{R_s^2}{D^2} f^2 T_d^4$$

$$T_D = T_s \left(\frac{d_1^2}{8f^2} \right)^{\frac{1}{4}} = 600 \left(\frac{10^2}{8(40\sqrt{2})^2} \right)^{\frac{1}{4}} = 1500 \text{ K}$$

3. Ans (10.00)

$$\Delta \ell = 2 \sqrt{\frac{1}{4} + (0.01)^2} - 1$$

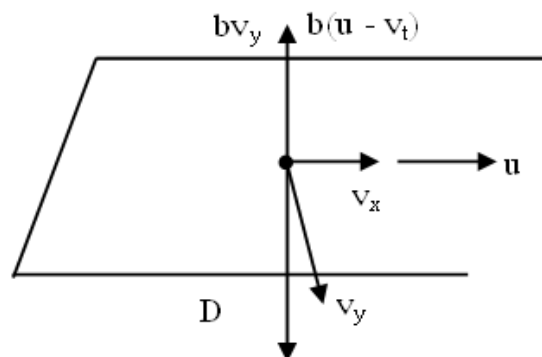
$$= 2 \times 10^{-4} \text{ m}$$

$$k \Delta \ell = 100$$

$$= (1 + 4 \times 10^{-4})^{1/2} - 1 \text{ k} = 5 \times 10^5$$

$$U = \frac{1}{2} \times 5 \times 10^5 \times 2^2 \times 10^{-8} = 10 \text{ mJ}$$

4. Ans (37.5)



$$mg - bv_y = ma_y$$

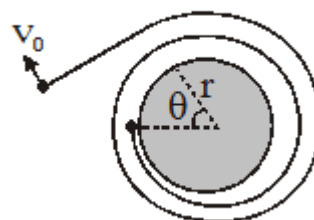
$$bu - bv_x = ma_x$$

$$a_y = g - \frac{bu}{m} \Rightarrow v_y = \frac{mg}{b} \left(1 - e^{-\frac{bt}{m}} \right)$$

$$a_x = \frac{bu}{m} - \frac{b}{m} u_y \Rightarrow v_x = u \left(1 - e^{-\frac{bt}{m}} \right)$$

$$\frac{y}{x} = \frac{mg}{ub} \Rightarrow x = \frac{D u b}{mg}$$

5. Ans (129.6)



$$-dl = r d\theta$$

$$\frac{d\theta}{dt} = \frac{v_0}{l}$$

$$-\int_L^0 l dl = rv_0 \int_0^t dt \Rightarrow \frac{L^2}{2} = rv_0 t$$

$$\text{time for unwinding } t_1 = \frac{L^2}{2rv_0}$$

then particle will move in a semicircle of radius

$$L \rightarrow t_2 = \frac{\pi L}{v_0}$$

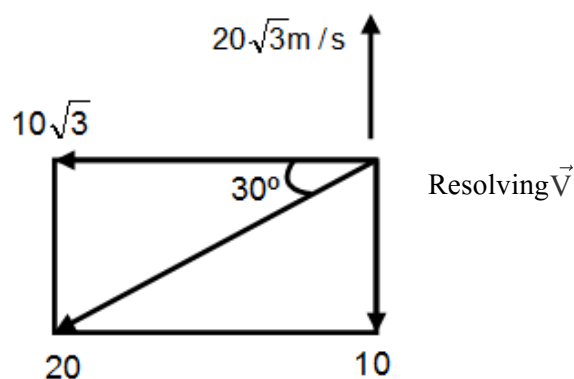
$$\text{then time for winding } t_3 = \frac{L^2}{2rv_0}$$

$$T = \frac{L^2}{rv_0} + \frac{\pi L}{v_0} = \frac{L}{rv_0} (L + \pi r)$$

6. **Ans (45.00)**

From cart's frame, initial velocity is $20\sqrt{3}\hat{j}$.

Constant acceleration of ball w.r.t. Cart = $-10\hat{j}$
 $-10\sqrt{3}\hat{i}$



as per constant $\vec{a}$

At point of minimum curvature, $10\sqrt{3} \text{ m/s}$ will become zero because of acceleration. At this instant acceleration is perpendicular to remaining velocity of 30 m/s

$$R_C = \frac{v^2}{a_n} = \frac{30^2}{20} = 45 \text{ m}$$

7. **Ans (18.00)**

$$\frac{dQ}{dt} = \frac{500 \times 5 \times 10^{-4} (T - 0)}{0.5}$$

$$\therefore Q = \int_0^{1200} \frac{500 \times 5 \times 10^{-4}}{0.5} \times \frac{t}{20} dt = 18 \text{ kJ}$$

8. **Ans (4.00)**

$$\vec{E} = \frac{a}{(x^2 + y^2 + z^2)} \frac{x\hat{i} + y\hat{j} + z\hat{k}}{\sqrt{x^2 + y^2 + z^2}} = \frac{a}{r^2} \hat{r}$$

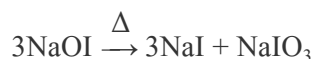
$$\text{for } \vec{E} = \frac{g}{4\pi\epsilon_0 r^2} \hat{r} \rightarrow \phi = \frac{q}{\epsilon_0}$$

$$\text{so for } \vec{E} = \frac{a}{r^2} \hat{r} \rightarrow \phi = \frac{4\pi\epsilon_0 a}{\epsilon_0} = 4\pi a$$

PART-2 : CHEMISTRY

SECTION-I(i)

1. **Ans (A,C,D)**



2. **Ans (A,B,C,D)**

For final solution

$$\frac{P^\circ - P_s}{P_s} = \frac{n}{N} \Rightarrow \frac{760 - 740}{740} = \frac{1}{N}$$

Moles of water left in solution = 37 mole

$$\text{RLVP of final solution} = \frac{n}{n+N} = \frac{1}{1+37} = \frac{1}{38}$$

$$(\Delta T_t)_{\text{original solution}} = \frac{2 \times \frac{1}{200 \times 18}}{1000}$$

$$(\Delta T_t)_{\text{original solution}} = -\frac{10}{18}^\circ \text{C}$$

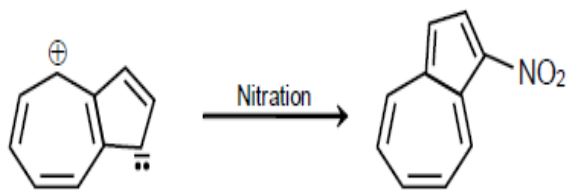
$$(\Delta T_t)_{\text{final solution}} = \frac{2 \times \frac{1}{37 \times 18}}{1000}$$

$$T = \left(273 - \frac{2000}{37 \times 18} \right) \text{K}$$

3. **Ans (A,C,D)**

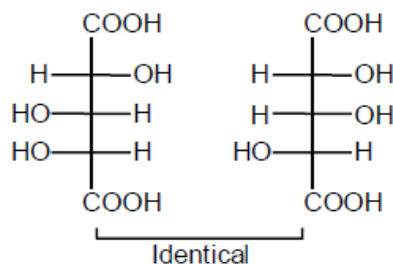
Formation of vacancy leads to 'missing bonds' and distortion of the lattice. The potential energy (internal energy & Enthalpy) of the system increases. Though it costs energy to form a vacancy, its formation leads to increase in configurational entropy (the crystal without vacancies represents just one state while the crystal with vacancies can exist in many energetically equivalent states, corresponding to various positions of the vacancies in the crystal).

5. Ans (A,B,C)



6. Ans (A,B,D)

(C) On switching we get the identical compound.



SECTION-I(ii)

7. Ans (B)

$$[H^+] \text{ in solution} = 2 \times 10^{-5} \times \frac{0.5}{0.1} = 10^{-4}$$

$$[OH^-] = \frac{10^{-14}}{10^{-4}} = 10^{-10}$$

$$\begin{aligned} \text{IP of Fe(OH)}_2 &= [Fe^{2+}][OH^-]^2 \\ &= \frac{0.05}{127} \times (10^{-10})^2 \\ &< K_{SP} \text{ of Fe(OH)}_2 \end{aligned}$$

∴ No precipitation of Fe(OH)₂

$$[S^{2-}] \text{ in solution} = K_a K_{a2} \frac{[H_2S]}{[H^+]^2} = \frac{10^{-21} \times (0.1)}{(10^{-4})^2} = 10^{-14} M$$

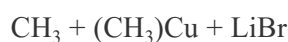
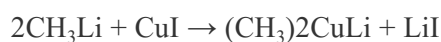
$$\text{IP of FeS} = [Fe^{2+}][S^{2-}] = \frac{0.05}{127} \times 10^{-14}$$

> K_{SP} of FeS

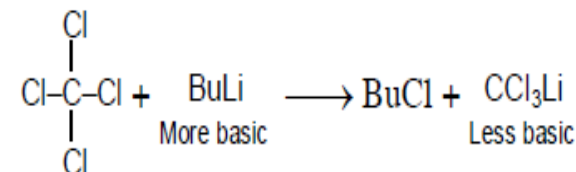
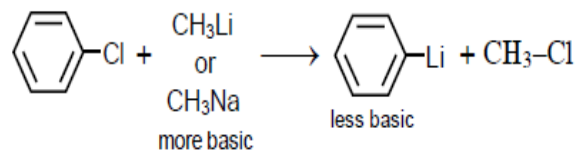
∴ Some FeS will precipitate out

$$\text{Finally } [Fe^{2+}] = \frac{8 \times 10^{-19}}{10^{-14}} = 8 \times 10^{-5} M$$

9. Ans (D)



10. Ans (C)



SECTION-II

1. Ans (3.00)

(Li⁺, Mg²⁺), (Li⁺, Al³⁺), (Li⁺_(aq), Na⁺_(aq))

2. Ans (5.00)

	P(g)	⇌	R(g) + S(g)	(i)
Initial pressure	100 cm of Hg			
Pressure of equilibrium	100-x		x	x
100+x = 150 or x = 50 cm of Hg	$K_{p1} = \frac{50 \times 50}{50} = 50$			
	Q(g)	⇌	2R(g) + T(g)	(ii)
Initial pressure	100		0	0
Pressure of equilibrium	100-y		2y	y

$$100+2y = 150 \text{ or } 2y = 50$$

$$K_{p2} = \frac{(50)^2 \times 25}{75}$$

In both equilibria partial pressure of R before connection is 50 cm of Hg, so

partial pressure of R after connection will be 50 cm of Hg.

$$Q_p \text{ for (i)} = \frac{50 \times \frac{50}{2}}{\frac{50}{2}} = K_{p1}$$

$$Q_p \text{ for (ii)} = \frac{(50)^2 \times \frac{25}{2}}{\frac{75}{2}} = K_{p2}$$

As both reaction are still at equilibrium, there will be no change in partial pressure of R.

4. Ans (4.00)

(ii) $\Delta H = \Delta U$ ($\Delta n_g = 0$)

$\Delta H = \Delta U = q_v$

(iii) For any irreversible isothermal expansion

$\Delta S_{\text{universe}} > 0$

$\Delta S_{\text{system}} + \Delta S_{\text{surr}} > 0$

$\Delta S_{\text{system}} \Rightarrow +ve$

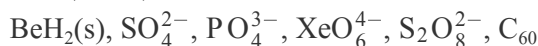
$\Delta S_{\text{surrounding}} \Rightarrow -ve \therefore |\Delta S_{\text{system}}| > |$

$\Delta S_{\text{surrounding}}|$

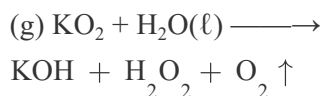
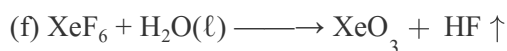
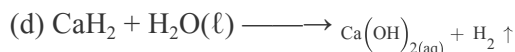
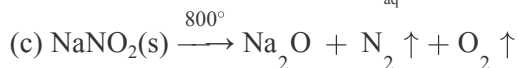
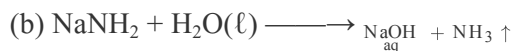
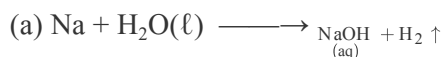
(vii) P (black, solid) $\longrightarrow$ P (white, solid) $\Delta S =$ positive

(viii) For mixing equal values of two ideal gases at constant temperature and pressure in an isolated container, $\Delta S_{\text{system}} > 0$ and $\Delta G < 0$

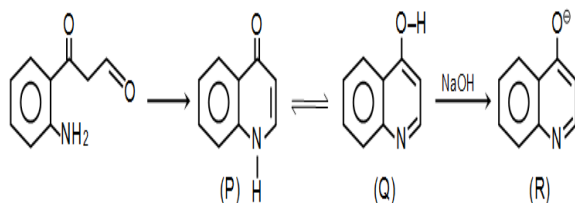
5. Ans (6.00)



6. Ans (7.00)

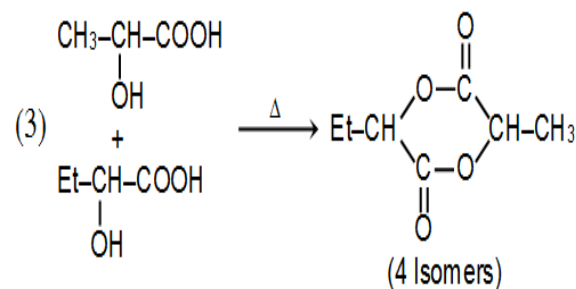
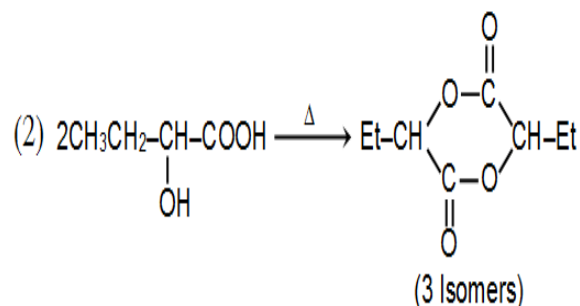
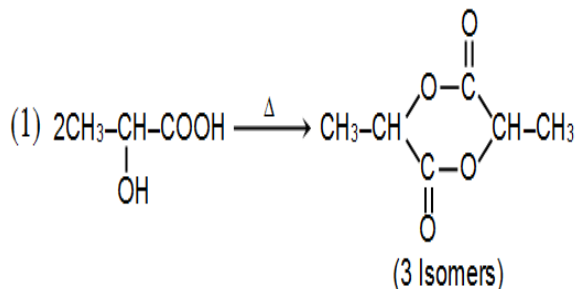
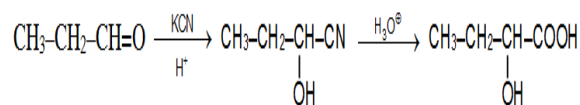
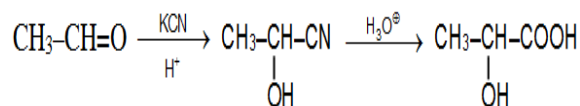


7. Ans (10.00)



(R) will have 10 resonating structures.

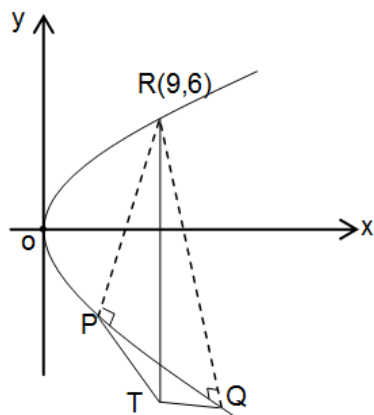
8. Ans (10.00)



PART-3 : MATHEMATICS

SECTION-I(i)

1. Ans (C,D)



parameter of R is $t_R = 3$

If $t_p t_q = 2$ & $t_p + t_q = -3$.

$\Rightarrow t_p = -1$ & $t_q = -2$ (or vice versa)

$\therefore P \equiv (1, -2)$ & $Q \equiv (4, -4)$.

$\therefore R \equiv (9, 6)$, $Q \equiv (4, -4) \Rightarrow T \equiv (2, -3)$ A ($\square$

PTQR) = PR $\times$ PT = 16

Equation of circle circumscribing ($\square$ PTQR) is : $x^2 + y^2 - 11x - 3y = 0$.

2. Ans (B,C,D)

$$(1 + x + x^2)^{100} = a_0 + a_1 x + \dots + a_{35} x^{35} + a_{36} x^{36} + a_{37} x^{37} + \dots + a_{200} x^{200}$$

Diff. both sides wrt x , we get :

$$100(1 + x + x^2)^{99}(2x + 1) = a_1 + 2a_2 x + \dots + 35a_{35} x^{34} + 36a_{36} x^{35} + 37a_{37} x^{36} + \dots$$

Multiply both sides by $(1 + x + x^2)$,

$$100(2x + 1)(1 + x + x^2)^{100} = (1 + x + x^2)(a_1 + \dots + 35a_{35} x^{34} + \dots)$$

$$\therefore 100(2x + 1)(a_0 + a_1 x + \dots + a_{200} x^{200}) = (1 + x + x^2)(a_1 + 2a_2 x + \dots + 35a_{35} x^{34} + \dots)$$

Comparing co-efficient of x^{36} on both sides, we get desired relation.

$$100(a_{36} + 2a_{35}) = (37a_{37} + 36a_{36} + 35a_{35})$$

3. Ans (A,B,C)

$$\frac{d^2 y}{dx^2} + \frac{2}{1-x} \frac{dy}{dx} + \frac{2y}{(1-x)^2} = 0$$

$$\Rightarrow \frac{d^2 y}{dx^2} + \frac{1}{1-x} \frac{dy}{dx} + \frac{1}{1-x} \frac{dy}{dx} + \frac{2y}{(1-x)^2} = 0$$

$$\Rightarrow \frac{1}{1-x} \frac{d^2 y}{dx^2} + \frac{1}{(1-x)^2} \frac{dy}{dx} + \frac{1}{(1-x)^2} \frac{dy}{dx} + \frac{2y}{(1-x)^3} = 0$$

$$\Rightarrow \frac{d}{dx} \left(\frac{1}{1-x} \frac{dy}{dx} \right) + \frac{d}{dx} \left(\frac{1}{(1-x)^2} y \right) = 0$$

On integrating

$$\frac{1}{1-x} \frac{dy}{dx} + \frac{y}{(1-x)^2} = C_1 \text{ at } x=0, y=1,$$

$$\frac{dy}{dx} = 0 \Rightarrow C_1 = 1$$

$$\Rightarrow \frac{1}{1-x} \frac{dy}{dx} + \frac{y}{(1-x)^2} = 1$$

$$\Rightarrow \frac{d}{dx} \left(\frac{1}{1-x} y \right) = 1$$

Integrating, we get :

$$\frac{y}{1-x} = x + C_2$$

$$\text{at } x=0, y=1 \Rightarrow C_2 = 1$$

$$\Rightarrow \frac{y}{1-x} = x + 1$$

$$\Rightarrow y = 1 - x^2$$

$$y(4) = -15$$

$$y = |f(x)| = |1-x^2| \quad \left(\begin{array}{c} \text{Graph of } y = |1-x^2| \text{ is shown, a downward parabola with vertex at } (0, 1) \text{ and x-intercepts at } \pm 1. \end{array} \right) |f(x)|$$

$|f(x)| = \frac{1}{2}$ has 4 solution

As $x = \pm 1$, concavity changes, but tangents can't be drawn so these cannot be point of inflection.

4. Ans (A,C)

$$\text{Here } h(x) = \frac{1}{1-x}; x \neq 1$$

$$\Rightarrow h(h(x)) = \frac{x-1}{x}; x \neq 0, 1$$

$$\therefore h(h(h(x))) = x; x \neq 0, 1$$

Also $g(x) \geq 0 \quad \forall x \in \mathbb{R}$

$$\Rightarrow f(x) \geq x$$

$$\Rightarrow f(f(x)) \geq f(x) \geq x$$

$$\Rightarrow f(f(f(x))) \geq f(x) \geq x$$

$$\Rightarrow f(f(f(x))) - x \geq 0$$

$$\Rightarrow f(f(f(x))) - h(h(h(x))) \geq 0$$

such that $x \neq 0, 1$

5. Ans (A,B,C)

$$p\alpha^{n+2} + q\beta^{n+2} - 3p\alpha^{n+1} - 3q\beta^{n+1} + 2p\alpha^n + 2q\beta^n = 0$$

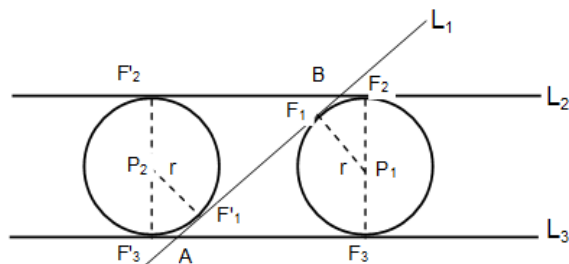
$$\Rightarrow p\alpha^n(\alpha^2 - 3\alpha + 2) + q\beta^n(\beta^2 - 3\beta + 2) = 0$$

$$\Rightarrow \alpha = 1, 2 \text{ and } \beta = 1, 2$$

$$\Rightarrow \text{equation can be } (x-1)^2 = 0 \text{ or } (x-2)^2 = 0$$

$$\text{or } (x-1)(x-2) = 0$$

6. Ans (A,D)



$$F_2F_2' = P_1P_2 = \sqrt{(\alpha - \gamma)^2 + (\beta - \delta)^2}$$

$$\text{Let } \angle P_1AF_3 = \theta \Rightarrow AF_3 = r \cot \theta = AF_1$$

$$\text{and } BF_2 = r \tan \theta = BF_1 = AF_1'$$

$$\Rightarrow F_1F_1' = |AF_1 - AF_1'| = r |\cot \theta - \tan \theta| = 2r |\cot 2\theta|$$

$$= 2r \cot 2\theta \text{ (because } 2\theta \text{ is acute angle between } L_1 \text{ and } L_3)$$

$$\text{Now } F_2F_2' = |r \cot \theta + r \tan \theta| = 2r |\csc 2\theta|$$

$$\Rightarrow (F_2F_2')^2 - (F_1F_1')^2 = (2r)^2 = \frac{\left| \frac{c_3}{a_3} - \frac{c_2}{a_2} \right|^2}{1^2 + \left(\frac{b_2}{a_2} \right)^2} =$$

$$\frac{(a_2c_3 - a_3c_2)^2}{a_3^2(a_2^2 + b_2^2)^2}$$

SECTION-I(ii)

7. Ans (D)

$$g(x) = x + \frac{1^2x^3}{3!} + \frac{1^23^2x^5}{5!} + \frac{1^23^25^2x^7}{7!} + \dots = \sin^{-1}x$$

$$\Rightarrow f(x) = \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow f\left(\frac{4}{5}\right) = \frac{5}{3}$$

$$\Rightarrow \int_0^1 g(x) dx = \int_0^1 \sin^{-1}x dx = \frac{\pi}{2} - 1$$

8. Ans (C)

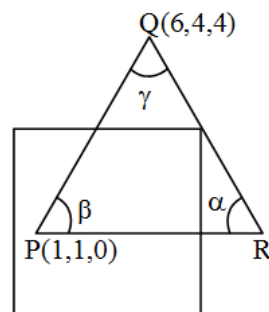
$$g(x) = x + \frac{1^2x^3}{3!} + \frac{1^23^2x^5}{5!} + \frac{1^23^25^2x^7}{7!} + \dots = \sin^{-1}x$$

$$\Rightarrow f(x) = \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow f\left(\frac{4}{5}\right) = \frac{5}{3}$$

$$\Rightarrow \int_0^1 g(x) dx = \int_0^1 \sin^{-1}x dx = \frac{\pi}{2} - 1$$

9. Ans (A)



$$\text{If } \frac{QP + PR}{QR} \text{ is maximum then } \frac{\sin \alpha}{\sin \beta} + \frac{\sin(\pi - (\alpha + \beta))}{\sin \beta} \text{ is maximum}$$

$$\frac{\sin \alpha}{\sin \beta} + \frac{\sin(\alpha + \beta)}{\sin \beta} = \frac{2 \sin\left(\frac{2\alpha + \beta}{2}\right) \cos \frac{\beta}{2}}{\sin \beta} = \frac{\sin\left(\alpha + \frac{\beta}{2}\right)}{\sin \frac{\beta}{2}}$$

$$\Rightarrow \text{If } \frac{PQ + PR}{QR} \text{ is maximum then}$$

$$\alpha + \frac{\beta}{2} = \frac{\pi}{2} \text{ and } \frac{\beta}{2} \text{ is minimum positive angle belongs to } \left(0, \frac{\pi}{2}\right)$$

$$\Rightarrow \alpha = \gamma \text{ and R lies on projection of PQ on plane.}$$

$$\text{Now foot of Q on plane is } S(2, 0, 0)$$

$$\text{Direction cosine of } \overrightarrow{PS} \text{ is}$$

$$\left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 0\right), \text{ Now } PQ = \sqrt{50} = PR$$

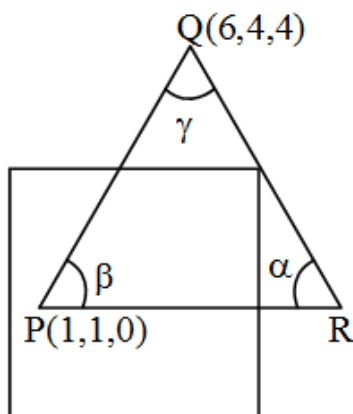
$$\text{coordinates of R is given by}$$

$$\frac{x-1}{\frac{1}{\sqrt{2}}} = \frac{y-1}{\frac{-1}{\sqrt{2}}} = \frac{z-0}{0} = \sqrt{50}$$

$$\Rightarrow x = 6, y = -4, z = 0 \Rightarrow R \text{ is } (6, -4, 0)$$

$$\bullet \ell = \frac{\sqrt{50} + \sqrt{50}}{\sqrt{80}} = \sqrt{\frac{5}{2}} \Rightarrow 2\ell^2 = 5$$

10. Ans (B)



If $\frac{QP + PR}{QR}$ is maximum then $\frac{\sin \alpha}{\sin \beta} + \frac{\sin(\pi - (\alpha + \beta))}{\sin \beta}$ is maximum Now

$$\frac{\sin \alpha}{\sin \beta} + \frac{\sin(\alpha + \beta)}{\sin \beta} = \frac{2 \sin\left(\frac{2\alpha + \beta}{2}\right) \cos \frac{\beta}{2}}{\sin \beta} = \frac{\sin\left(\alpha + \frac{\beta}{2}\right)}{\sin \frac{\beta}{2}}$$

$\Rightarrow$ If $\frac{PQ + PR}{QR}$ is maximum then

$\alpha + \frac{\beta}{2} = \frac{\pi}{2}$ and $\frac{\beta}{2}$ is minimum positive angle belongs to $\left(0, \frac{\pi}{2}\right)$

$\Rightarrow \alpha = \gamma$ and R lies on projection of PQ on plane.

Now foot of Q on plane is S(2, 0, 0)

Direction cosine of $\overrightarrow{PS}$ is

$$\left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 0\right), \text{ Now } PQ = \sqrt{50} = PR$$

coordinates of R is given by

$$\frac{x-1}{\frac{1}{\sqrt{2}}} = \frac{y-1}{\frac{-1}{\sqrt{2}}} = \frac{z-0}{0} = \sqrt{50}$$

$\Rightarrow x = 6, y = -4, z = 0 \Rightarrow R$ is (6, -4, 0)

$$\bullet \ell = \frac{\sqrt{50} + \sqrt{50}}{\sqrt{80}} = \sqrt{\frac{5}{2}} \Rightarrow 2\ell^2 = 5$$

SECTION-II

1. Ans (1.00)

$$\begin{aligned} g(y) &= \sum_{n=1}^{\infty} \lim_{x \rightarrow 0} (-y)^n \left(\frac{f\left(\frac{x}{n}\right) - f(0)}{\frac{x}{n}} \right) \times \frac{1}{n} \\ &= \sum_{n=1}^{\infty} \frac{(-y)^n f'(0)}{n} = \sum_{n=1}^{\infty} \frac{(-y)^n}{n} = -\ln(1+y) \\ \Rightarrow g(-y) &= -\ln(1-y) \Rightarrow g(y) + g(-y) = -\ln((1-y)(1+y)) = -\ln(1-y^2) = g(-y^2) \\ \Rightarrow \frac{g(y) + g(-y)}{g(-y^2)} &= 1 \end{aligned}$$

2. Ans (2.50)

Because discriminant of $x^2 + (2-a)x + 3 = 0$ is negative so $x^2 + (2-a)x + 3 = 0$ has both imaginary roots

$\Rightarrow$ both the equation have two common roots

Now, roots of equation $x^2 + (2-a)x + 3 = 0$ are roots of

$$(ax^4 + bx^3 + x^2 + (3-a)x + 3) - (x^2 + (2-a)x + 3) = 0$$

$\Rightarrow$ roots of equation $x^2 + (2-a)x + 3 = 0$ are roots of $ax^4 + bx^3 + x = 0$

$\Rightarrow$ roots of equation $x^2 + (2-a)x + 3 = 0$ are roots of $ax^3 + bx^2 + 1 = 0$

$\Rightarrow$ roots of equation $x^2 + (2-a)x + 3 = 0$ are roots of $(ax^3 + bx^2 + 1) - (ax(x^2 + (2-a)x + 3)) = 0$

$\Rightarrow$ roots of equation $x^2 + (2-a)x + 3 = 0$ and $(b - 2a + a^2)x^2 - 3ax + 1 = 0$ are same

$$\Rightarrow \frac{1}{b - 2a + a^2} = \frac{2-a}{-3a} = \frac{3}{1}$$

$$\Rightarrow a = -\frac{1}{4} \text{ and } b = -\frac{11}{48}$$

$$\Rightarrow |a - 12b| = 2.5$$

3. Ans (4.00)

$$\begin{aligned} \text{Let } g(\alpha) &= \int_{-2}^{\alpha} (x+2)(x-2)(x-\alpha) dx \\ &= \int_{-2}^{\alpha} (x+2)(x-2)x dx - \alpha \int_{-2}^{\alpha} (x+2)(x-2) dx \\ g'(\alpha) &= (\alpha+2)(\alpha-2)\alpha - \alpha(\alpha+2)(\alpha-2) - \int_{-2}^{\alpha} (x+2)(x-2) dx \\ &= -\left[\frac{x^3}{3} - 4x\right]_{-2}^{\alpha} \\ &= -(\alpha+2)^2(\alpha-4) = 0 \Rightarrow \alpha = -2, 4 \\ \text{at } \alpha = 4, A \text{ is maximum} \end{aligned}$$

4. Ans (0.25)

Let point of intersection of curves is (x_1, y_1)

$$\text{and let } 2 \cot \sqrt{\frac{\pi^2}{16} - \alpha^2} = k$$

$$\text{Now differentiate } \log(x^2 + y^2) = k \tan^{-1} \frac{y}{x}$$

w.r.t. x we get

$$\begin{aligned} \frac{1}{x^2 + y^2} \left(2x + 2y \frac{dy}{dx} \right) &= k \left(\frac{1}{1 + \frac{y^2}{x^2}} \right) \\ \left\{ \frac{x \frac{dy}{dx} - y}{x^2} \right\} \end{aligned}$$

$$\Rightarrow 2 \left\{ x + y \frac{dy}{dx} \right\} = k \left\{ x \frac{dy}{dx} - y \right\}$$

$$\Rightarrow \frac{dy}{dx} = \frac{2x + ky}{kx - 2y}$$

$$\Rightarrow \left. \frac{dy}{dx} \right|_{(x_1, y_1)} = \frac{2x_1 + ky_1}{kx_1 - 2y_1} = m_1$$

$$\frac{dy}{dx} \text{ for } y = mx \text{ is } m = \frac{y_1}{x_1}$$

$$\Rightarrow \tan \theta = \left| \frac{\frac{2x_1 + ky_1}{kx_1 - 2y_1} - \frac{y_1}{x_1}}{1 + \left(\frac{2x_1 + ky_1}{kx_1 - 2y_1} \right) \left(\frac{y_1}{x_1} \right)} \right| = \left| \frac{2x_1^2 + kx_1y_1 - kx_1y_1 + 2y_1^2}{kx_1^2 - 2x_1y_1 + 2x_1y_1 + ky_1^2} \right| = \left| \frac{2}{k} \right|$$

$$\Rightarrow \tan \theta = \tan \sqrt{\frac{\pi^2}{16} - \alpha^2}$$

$$\Rightarrow \theta = \sqrt{\frac{\pi^2}{16} - \alpha^2}$$

$$\Rightarrow \text{max value of } \theta \text{ is } \frac{\pi}{4}$$

$$\Rightarrow b = \frac{1}{4} = 0.25$$

5. Ans (156.00)

$$\begin{vmatrix} a & b \\ d & e \end{vmatrix} = 0 \Rightarrow \frac{a}{d} = \frac{b}{e} \text{ and}$$

$$\begin{vmatrix} b & c \\ e & f \end{vmatrix} = 0 \Rightarrow \frac{b}{e} = \frac{c}{f}$$

$$\Rightarrow \frac{a}{d} = \frac{b}{e} = \frac{c}{f}$$

$$\text{similarly } \frac{d}{g} = \frac{e}{h} = \frac{f}{i} \text{ and } \frac{a}{g} = \frac{b}{h} = \frac{c}{i}$$

$$a : b : c = d : e : f = g : h : i$$

Case-I $\{a, b, c\} = \{d, e, f\} = \{g, h, i\}$ then total

$$\text{matrices possible equals to } {}^6C_3 \times 3! = 120$$

Case-II two rows the same and one different

$$\text{e.g. } \{a, b, c\} = \{1, 2, 3\} \text{ and } \{d, e, f\} = \{g, h, i\}$$

$$= \{2, 4, 6\}$$

$$\text{the total matrices possible equals to } {}^3C_2 \times 3! \times$$

$$2 = 36$$

$$\text{total } 156$$

6. Ans (3.00)

$$k = \lim_{x \rightarrow \infty} (x+2) \tan^{-1}(x+2) - x \tan^{-1} x$$

$$= \lim_{x \rightarrow \infty} x (\tan^{-1}(x+2) - \tan^{-1} x) + 2 \tan^{-1}(x+2)$$

$$= \lim_{x \rightarrow \infty} x \left[\tan^{-1} \left(\frac{2}{1 + x(x+2)} \right) \right] + 2 \tan^{-1}(x+2)$$

$$= \lim_{x \rightarrow \infty} x \tan^{-1} \left(\frac{2}{(x+1)^2} \right) + 2 \tan^{-1}(x+2)$$

$$= \lim_{x \rightarrow \infty} \frac{\tan^{-1} \left(\frac{2}{(x+1)^2} \right)}{\frac{2}{(x+1)^2}} \times \frac{2x}{(x+1)^2} + 2 \tan^{-1}(x+2)$$

$$k = 1 \times 0 + 2 \times \frac{\pi}{2} = \pi \Rightarrow k = \pi \quad \therefore [k] = 3$$

7. Ans (1.00)

$$y = \lim_{n \rightarrow \infty} \left[\frac{(n+x)(n+2x)\dots(n+2nx)}{n^{2n}} \right]^{\frac{1}{2n}} \cdot \left(\tan \frac{\pi}{2n} \cdot \tan \frac{2\pi}{2n} \dots \right)^{\frac{1}{n}}$$

$$= \lim_{n \rightarrow \infty} \left[\left(1 + \frac{x}{n} \right) \left(1 + \frac{2x}{n} \right) \dots \left(1 + \frac{2nx}{n} \right) \right]^{\frac{1}{2n}} \times \prod_{r=1}^{n-1} \left[\tan \left(\frac{r\pi}{2n} \right) \right]^{\frac{1}{n}}$$

$$= \lim_{n \rightarrow \infty} \prod_{r=1}^{2n} \left(1 + \frac{rx}{n} \right)^{\frac{1}{2n}} \prod_{r=1}^{n-1} \left(\tan \frac{r\pi}{2n} \right)^{\frac{1}{n}}$$

$$\therefore \ln y$$

$$= \lim_{n \rightarrow \infty} \left\{ \frac{1}{2n} \sum_{r=1}^{2n} \ln \left(1 + \frac{rx}{n} \right) + \frac{1}{n} \sum_{r=1}^{n-1} \ln \left(\tan \frac{r\pi}{2n} \right) \right\}$$

$$\ln y =$$

$$\frac{1}{2} \int_0^2 \ln(1+tx) dt + \int_0^1 \ln \left(\tan \frac{t\pi}{2} \right) dt$$

$$= \frac{1}{2x} \int_0^{2x} \ln(1+t) dt + \frac{2}{\pi} \int_0^{\pi/2} \ln \tan t dt$$

$$= \frac{1}{2x} [(1+t)\ln(1+t) - t]_0^{2x} + \frac{2}{\pi} \int_0^{\pi/2} (\ln \sin t - \ln \cos t) dt$$

$$= \frac{1}{2x} [(1+2x)\ln(1+2x) - 2x] + \frac{2}{\pi} (0)$$

$$\Rightarrow \lim_{x \rightarrow 0} \ln y = \lim_{x \rightarrow 0} \frac{(1+2x)\ln(1+2x)}{2x} - 1 = 0$$

$$\Rightarrow \lim_{x \rightarrow 0} y = \lim_{x \rightarrow 0} e^{\ln y} = e^0 = 1$$

8.

Ans (190.00)

Let the number of different coloured balls = x

Let the number of balls of each colour = y

Now, P(getting two balls of same colour) =

$$\frac{xy(y-1)}{xy(xy-1)} \dots (1)$$

After adding 20 balls

$$P(\text{getting same colour}) = \frac{xy(y-1) + 20.19}{(xy+20)(xy+19)} \dots (2)$$

From (1) and (2)

$$\frac{xy(y-1)}{xy(xy-1)} = \frac{xy(y-1) + 20.19}{(xy+20)(xy+19)}$$

Let $xy = A$ and $y-1 = B$

$$\Rightarrow A = \frac{19(B+1)}{19-2B} \Rightarrow B = 9, A = 19 \times 10$$

$$y = 10, xy = 19 \times 10$$

$$\Rightarrow x = 19$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Advanced)

Full Syllabus

31-07-2021

JEE(Main + Advanced) : Leader & Enthusiast Course (PHASE : SCORE-3.0)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I(i)	Q.	1	2	3	4	5	6		
	A.	A,C	A,C	A,D	A,C	B	A,C		
SECTION-I(ii)	Q.	7	8	9	10				
	A.	B	C	D	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	22.00	0.50	43.50	4.00	8.00	20.00	5.00	1.80

PART-2 : CHEMISTRY

SECTION-I(i)	Q.	1	2	3	4	5	6		
	A.	A,B,C	B,C	A,B,C	C	A,C	A,B,C,D		
SECTION-I(ii)	Q.	7	8	9	10				
	A.	B	A	C	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	1.11	9.00	10.00	7.00	4.00	4.00	25.00	8.00

PART-3 : MATHEMATICS

SECTION-I(i)	Q.	1	2	3	4	5	6		
	A.	A,C	A,B	A,B,C,D	A,C,D	A,B,D	B,C		
SECTION-I(ii)	Q.	7	8	9	10				
	A.	A	B	C	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.00	7.00	5.00	173.00	3.00	0.25	7.00	8.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I(i)

1. Ans (A,C)

Conservation of angular momentum,

$$m_s a^2 \omega_A = (m_s + m_B) b^2 \omega'$$

and angular speed of A does not change because flying sand carries same angular momentum.

2. Ans (A,C)

$$u = \int_0^{R_1} \frac{1}{2} \epsilon_0 \left(\frac{\rho r}{3\epsilon_0} \right)^2 4\pi r^2 dr$$

$$= \frac{1}{2} \frac{\epsilon_0 \rho^2}{q \epsilon_0^2} 4\pi \frac{R_1^5}{5}$$

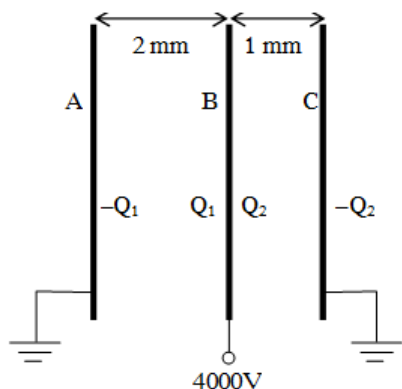
$$= \frac{2\pi \rho^2 R_1^5}{45\epsilon_0}$$

$$V_{\text{centre}} = \frac{3KQ}{2R_1} - \frac{KQ}{R_1} = \frac{KQ}{2R_1}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{\rho \times \frac{4}{3}\pi R_1^3}{2R_1}$$

$$= \frac{\rho R_1^2}{6\epsilon_0}$$

4. Ans (A,C)



$$Q_1 = \frac{\epsilon_0 A}{2d} V, Q_2 = \frac{\epsilon_0 A}{d} V$$

$$V_{AC} = \frac{(Q_2 - Q_1)3d}{2\epsilon_0 A} = \frac{\epsilon_0 A V}{2d} \cdot \frac{3d}{2\epsilon_0 A}$$

$$= \frac{3V}{4} = 3000 \text{ V}$$

$$\Delta U = \frac{1}{2} \epsilon_0 \left(\frac{3V}{4.3d} \right)^2 A 3d$$

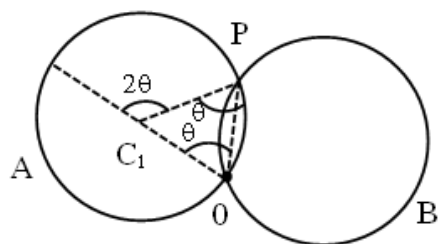
$$- \left[\frac{1}{2} \epsilon_0 \left(\frac{V}{2d} \right)^2 + \frac{1}{2} \epsilon_0 \left(\frac{V}{d} \right)^2 A d \right]$$

$$= \frac{3}{32} \frac{\epsilon_0 A}{d} V^2 - \left[\frac{3}{4} \frac{\epsilon_0 A}{d} V \right]$$

$$= -\frac{3 \times 7}{32} \frac{\epsilon_0 A}{d} V^2$$

$$= \frac{-3 \times 7}{32} \times \frac{8.85 \times 10^{-12} \times 2 \times 10^{-2}}{10^{-3}} \times (4000)^2$$

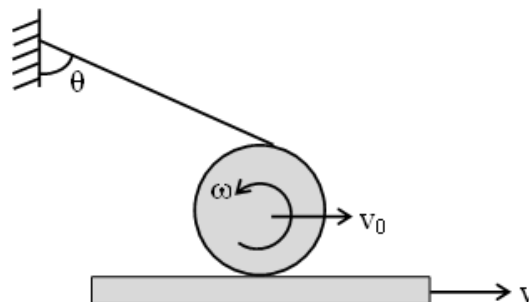
5. Ans (B)



$$\omega_{po} = \frac{d\theta}{dt} = -\omega \quad \omega_{pc_1} = \frac{d(2\theta)}{dt} = -2\omega$$

$$v_p = 2\omega R$$

6. Ans (A,C)



$$v_0 + \omega R = v$$

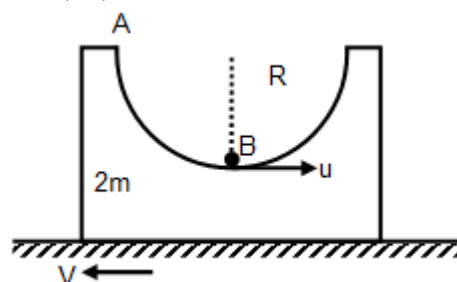
$$v_0 \sin \theta - \omega R = 0$$

$$v_0 = \frac{v}{1 + \sin \theta}$$

$$\frac{dv_0}{dt} \neq 0$$

SECTION-I(ii)

7. Ans (B)



Let the velocity of the mass m w.r.t. 2m be u, and the velocity of 2m w.r.t. ground be V. Then, applying conservation of mechanical energy at points A and B.

$$E = \frac{1}{2} m(u - V)^2 + \frac{1}{2} (2m)V^2 \dots (1)$$

conserving momentum in the horizontal direction.

$$0 = m(u - V) - 2mV \Rightarrow u = 3V \dots (2)$$

Solving (1) and (2) $mV^2 = \frac{E}{3}$. Also $mu^2 = 3E$

$$\therefore \text{KE of } 2m = \frac{1}{2} \times 2m \times V^2 = \frac{E}{3}$$

From (1) we get

$$E = \frac{1}{2} m(u - V)^2 + \frac{E}{3}$$

$$\Rightarrow \frac{1}{2} m(u - V)^2 = \frac{2E}{3} = \text{KE of } m$$

$$\text{Now, KE of } 2m = K_2 = \frac{P_2^2}{2(2m)} = \frac{E}{3}$$

$$\Rightarrow P_2 = \sqrt{\frac{4mE}{3}}$$

$$\text{KE of } m = K_1 = \frac{P_1^2}{2m} = \frac{2E}{3} \Rightarrow P_1 = \sqrt{\frac{4mE}{3}}$$

9. Ans (D)

$$\text{Maximum K.E loss} = \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} v^2$$

$$= \frac{1}{2} \left(\frac{1}{2} m v^2 \right) \text{ (as } m_n \simeq m_H \text{)}$$

$$= \frac{1}{2} \text{ K.E of neutron.}$$

$$(P) \text{ max K.E loss} = \frac{10.2}{2} = 5.1 \text{ eV}$$

So no excitation, hence no emission.

$$(Q) \text{ max K.E loss} = \frac{20.4}{2} = 10.2 \text{ eV}$$

Electron may get excited to 2nd shell, so 1 possible emission.

$$(R) \text{ max K.E loss} = \frac{24.2}{2} \text{ eV} = 12.1 \text{ eV}$$

e- excited to 3rd state, so 3 possible emission.

$$(S) \text{ max K.E loss} = \frac{25.51}{2} = 12.755 \text{ eV}$$

e- excited to 4th state, so 6 possible emission.

SECTION-II

1. Ans (22.00)

Pressure at point x

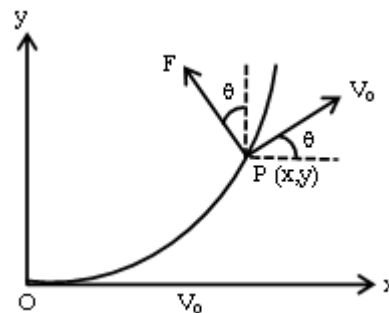
$$= P_0 + \frac{2T}{3R}$$

Pressure at point y

$$= P_0 + \frac{2T}{3R} + \frac{2T}{R} + \frac{4T}{R/3} = P_0 + \frac{44T}{3R}$$

$$\therefore \frac{(\text{Gauge pressure})_y}{(\text{Gauge pressure})_x} = \frac{44}{2} = 22$$

2. Ans (0.50)



Particle moves in xy plane. Its path is not a circle because B is not constant. However, speed of the particle remains constant at V_0 .

Let velocity at point P (x,y) be inclined to x axis at an angle θ .

Force will be perpendicular to velocity, inclined at θ with y direction.

$$\therefore m a_y = F \cos \theta$$

$$\Rightarrow m \frac{dV_y}{dt} = (B_0 x) q V_0 \cos \theta$$

$$\Rightarrow m \frac{dV_y}{dx} \cdot \frac{dx}{dt} = B_0 q x (V_x) [\because V_0 \cos \theta = V_x]$$

$$\frac{dx}{dt} = V_x$$

$$\text{But } \frac{dV_y}{dx} = \frac{B_0 q}{m} x$$

$$\Rightarrow \int_0^{V_0} dV_y = \frac{B_0 q}{m} \int_0^{x_0} x dx$$

Where x_0 = maximum x co-ordinate of the particle where velocity becomes parallel to y axis.

$$\therefore V_0 = \frac{B_0 q}{2m} x_0^2$$

$$\Rightarrow x_0 = \sqrt{\frac{2mV_0}{B_0 q}}$$

Magnetic force at $x = x_0$ is $F = q V_0 (B_0 x_0)$

$$\therefore F = q V_0 B_0 \sqrt{\frac{2mV_0}{B_0 q}} = v_0 \sqrt{2mqB_0 V_0}$$

If radius of curvature of path is R then

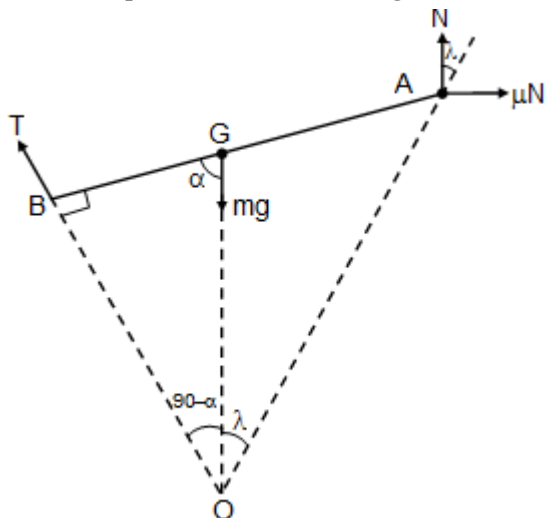
$$\frac{m V_0^2}{R} = F$$

$$\Rightarrow \frac{m V_0^2}{R} = v_0 \sqrt{2mqB_0 V_0}$$

$$\Rightarrow R = \sqrt{\frac{m V_0}{2qB_0}}$$

3. Ans (43.50)

For equilibrium, all force vectors should intersect at point O as shown in figure.



For ΔOAB

$$(BG + AG) \cot \alpha = AG \cot \lambda - BG \cot (90 - \alpha)$$

[m-n theorem]

$$\therefore L \cot \alpha = \frac{L}{2} \cot \lambda - \frac{L}{2} \tan \alpha$$

$$\therefore 2 \cot \alpha = \cot \lambda - \tan \alpha$$

$$\therefore \cot \lambda = \frac{2 + \tan^2 \alpha}{\tan \alpha}$$

$$\therefore \tan \lambda = \mu = \frac{\tan 37}{2 + \tan^2 37} = \frac{12}{41} = \frac{a}{b}$$

$$\text{Hence } \frac{3}{2}(b - a) = 43.50$$

4. Ans (4.00)

$$\text{Area of } \Delta ABE = \frac{1}{2} (2.5 \times 10^5) (0.05 \times 10^{-3})$$

$$= \frac{125}{2} = \frac{25}{4}$$

since, ΔAEB and ΔBCD are similar, therefore

$$\frac{1.5}{2.5} = \frac{CD}{0.05}$$

$$CD = \frac{3}{5} \times 0.05 = 0.03$$

Area of $\Delta BCD =$

$$\frac{1}{2} \times (1.5 \times 10^5) (0.03 \times 10^{-3})$$

$$= \frac{4.5}{2} = \frac{9}{4}$$

$$W_N = \frac{25}{4} - \frac{9}{4} = 4J$$

5. Ans (8.00)

$$\text{Initially, } I_1 = \frac{10}{10 + 1 + 6 + R/n} = \frac{10}{17 + R/n}$$

$$= 0.5 \text{ amp.(i)}$$

After removal of two resistors,

$$I_2 = \frac{10}{17 + R/(n-2)}$$

$$\frac{I_1 - I_2}{I_1} = \frac{100}{21} \times \frac{1}{100}$$

$$1 - \frac{I_2}{I_1} = \frac{1}{21}$$

$$I_2 = I_1 \times \frac{20}{21} = 0.5 \times \frac{20}{21} = \frac{10}{21} = \frac{10}{17 + \frac{R}{n-2}}$$

$$\text{or } \frac{R}{n-2} = 4, \text{ or } R = 4n - 8$$

$$\text{From (i), } \frac{0.5R}{n} + 8.5 = 10, R = 3n$$

$$\text{Hence, } 3n = 4n - 8$$

$$\text{or } n = 8$$

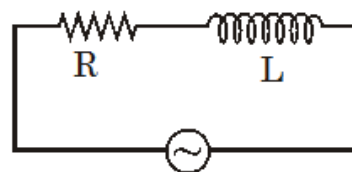
6. Ans (20.00)

Look at the given figure carefully to find that two plates of the capacitor marked as C are directly connected to the terminals of the battery. Potential difference across the marked capacitor is 20 V.

$$\therefore \text{Charge on it} = 20 \mu C$$

7. Ans (5.00)

Coil has internal resistance r.



$$40^2 = (ix_L)^2 + (ir)^2$$

$$50^2 = (ix_L)^2 + (i(r + 50))^2$$

$$i = \frac{20}{50} = 0.4$$

$$900 = 0.16[100r + 2500]$$

$$900 = 16r + 400$$

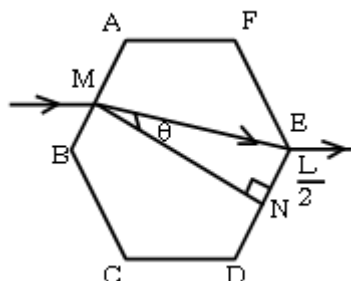
$$r = \frac{500}{16} \Omega$$

$$P = i^2 r$$

$$= 0.16 \times \frac{500}{16} = 5W$$

8. **Ans (1.80)**

For light beam to emerge parallel to the incident direction it must strike the face DE which is parallel to face AB. It means incident ray at M must go to E (or below it)
Let side length of hexagon be b.



$$MN = \sqrt{3}b$$

$$ME = \sqrt{\left(\frac{b}{2}\right)^2 + (\sqrt{3}b)^2} = \frac{\sqrt{13}}{2}b$$

Using Snell's law

$$1. \sin 30^\circ = \mu \sin \theta$$

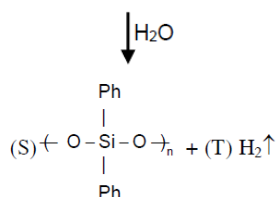
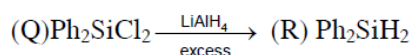
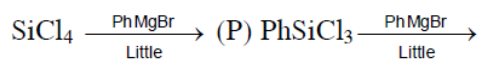
$$\Rightarrow \frac{1}{2} = \mu \frac{1}{\sqrt{3}}$$

$$\Rightarrow \mu = \frac{\sqrt{13}}{2}$$

PART-2 : CHEMISTRY

SECTION-I(i)

2. **Ans (B,C)**



3. **Ans (A,B,C)**

n-factor of MgCl_2 is 2. Thus, Limiting equivalent conductivity = $240/2 = 120 \text{ S.cm}^2.\text{eq}^{-1}$.

λ varies linearly with square root of concentration.

Put given values in equation to get value of constant A:

$$220 = 240 - A(0.01)^{0.5}$$

$$220 = 240 - A(0.1)$$

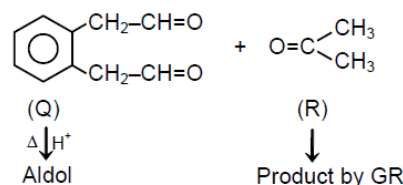
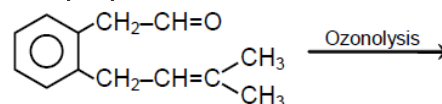
$$A = 200$$

Now, put $C = 0.04 \text{ M}$

$$\lambda = 240 - 200(0.04)^{0.5}$$

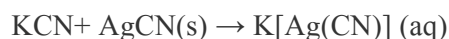
$$\lambda = 240 - 200(0.2) = 200 \text{ S.cm}^2.\text{mol}^{-1}$$

4. **Ans (C)**

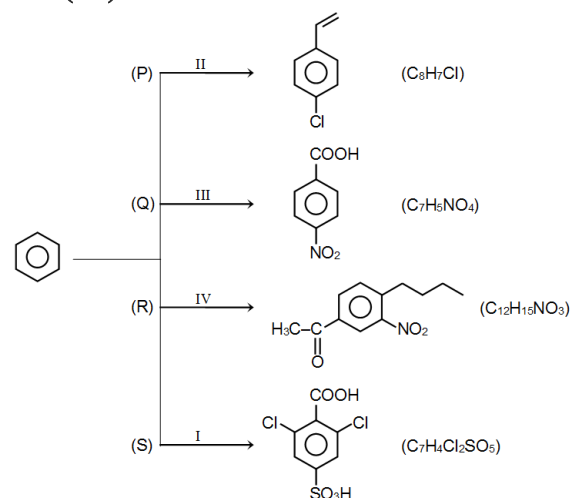


SECTION-I(ii)

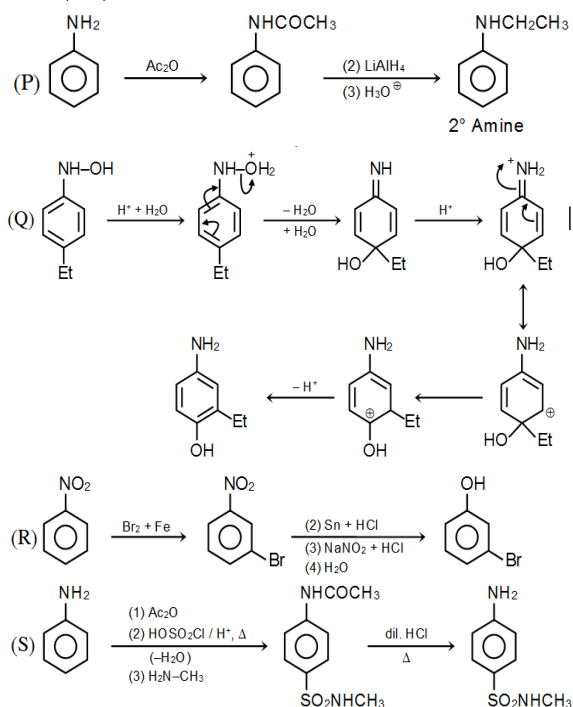
8. **Ans (A)**



9. **Ans (C)**



10. Ans (C)



SECTION-II

1. Ans (1.11)

$$Z = 1 + \frac{Pb}{RT}$$

$$\text{or } 1.11 - 1 = \frac{Pb}{RT}$$

$$\text{or } \frac{RT}{P} = \frac{b}{0.11} \text{ or } (V_m - b) = \frac{b}{0.11}$$

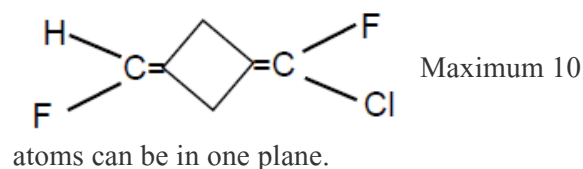
$$\text{or } V_m/b = 111/11$$

2. Ans (9.00)



$$\text{So, } 12 - 3 = 9.$$

3. Ans (10.00)



4. Ans (7.00)

(i) $s + p_x(x) = \sigma$ bond

(ii) $d_{xy} + d_{xy}(x) = \pi$ bond

(iii) $d_{yz} + p_z(z) = \text{no bond}$

(iv) $s + s(z) = \sigma$ bond

(v) $d_{yz} + d_{yz}(x) = \delta$ bond

(vi) $p_y + p_y(y) = \sigma$ bond

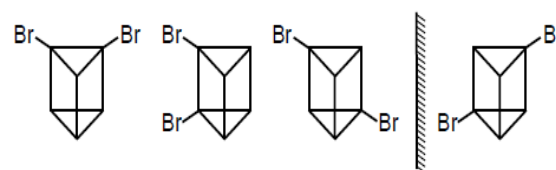
(vii) $p_z + p_z(z) = \text{no bond}$

(viii) $d_{xy} + d_{xy}(z) = \delta$ bond

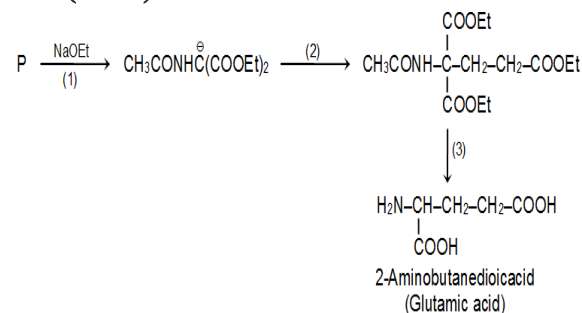
(ix) $p_x + p_x(y) = \pi$ bond

(x) $s + p_x(x) = \sigma$ bond

5. Ans (4.00)



6. Ans (4.00)



8. Ans (8.00)

$$E = \frac{1}{2} mV^2 \text{ where } E \text{ is kinetic energy}$$

$$dE = \frac{1}{2} m \cdot 2VdV + V^2 dm$$

$$\frac{dE}{E} = \frac{\frac{1}{2} m 2V dV + \frac{1}{2} V^2 dm}{\frac{1}{2} mV^2}$$

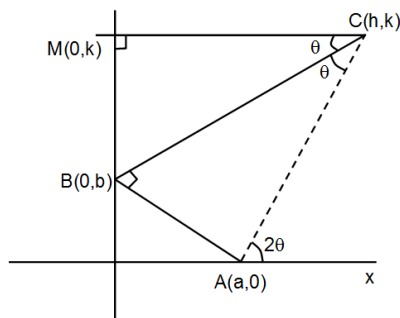
$$\frac{dE}{E} = 2 \frac{dV}{V} + \frac{dm}{m}$$

$$= 2 \times 3 + 2 = 8 \%$$

PART-3 : MATHEMATICS

SECTION-I(i)

1. **Ans (A,C)**



Let M is foot of C on y-axis, $\angle ACB = \theta$

$\Rightarrow \angle CAX = 2\theta$ (given), $\Rightarrow \angle MCA = 2\theta$

$\Rightarrow \angle MCB = \theta$

In $\triangle BMC$, $\tan \theta = \frac{k-b}{h}$ and In $\triangle BOA$, $\tan \theta$

$$= \frac{a}{b}$$

$$\text{In } \triangle BAC, \tan \theta = \frac{\sqrt{a^2 + b^2}}{\sqrt{h^2 + (k-b)^2}}$$

$$\Rightarrow \tan^2 \theta = \frac{a^2}{b^2} = \frac{b^2 + a^2}{h^2 + (k-b)^2} = \frac{(k-b)^2}{h^2}$$

$$= \frac{b^2}{h^2 + k^2 - 2bk} = \frac{k^2 - 2bk}{2bk - k^2} = \left(-1 \text{ or } \frac{0}{0}\right)$$

Because $\tan^2 \theta$ can not be equal to -1 so one possibility remaining is $k^2 = 2bk$

$$\Rightarrow k = 2b$$

$$\text{Put } k = 2b \text{ in } \frac{k-b}{h} = \frac{a}{b}$$

$$\Rightarrow k^2 = 4ah \Rightarrow y^2 = 4ax$$

2. **Ans (A,B)**

Equation of major axis of Ellipse is

$$\frac{x}{6} = \frac{y}{2} = \frac{z}{-3} \text{ and centre is } (0, 0, 0)$$

D.Rs of directrices are components of

$$(6\hat{i} + 2\hat{j} - 3\hat{k}) \times (2\hat{i} + 3\hat{j} + 6\hat{k}) \Rightarrow (3, -6, 2)$$

point of intersection of directrices and major

axis are $(150, 50, -75)$ and $(-150, -50, 75)$

$\Rightarrow$ equation of directrices are

$$\frac{x-150}{3} = \frac{y-50}{-6} = \frac{z+75}{2} \text{ and}$$

$$\frac{x+150}{3} = \frac{y+50}{-6} = \frac{z-75}{2}$$

3. **Ans (A,B,C,D)**

Point of intersection of circle $x^2 + y^2 = 4$ and

$(x + (\sqrt{3}+1))^2 + (y - (1 + \sqrt{3}))^2 = 8$ is $(-2, 0)$

and $(0, 2)$ if root are possible then they will be either -2 or $2i$

If $2i$ is root then $i^n = 1$ and $((1 + \sqrt{3}) + (1 -$

$$\sqrt{3}i)^n = (2\sqrt{2})^n$$

$$\Rightarrow n = 4k \text{ where } k \in \mathbb{I} \text{ and}$$

$$\left(((1 + \sqrt{3})^2 - (1 - \sqrt{3})^2) + 2(1 - 3)i \right)^{\frac{n}{2}} = (8)^{\frac{n}{2}}$$

$$\Rightarrow n = 4k, k \in \mathbb{I} \text{ and } (4\sqrt{3} - 4i)^{n/2} = (8)^{n/2}$$

$$\Rightarrow n = 4k, k \in \mathbb{I} \text{ and } (\sqrt{3} - i)^{n/2} = (2)^{n/2}$$

$$\Rightarrow n = 4k, k \in \mathbb{I} \text{ and } \frac{n}{2} \in 12\lambda, \lambda \in \mathbb{I}$$

$$\Rightarrow n \in 24\lambda, \lambda \in \mathbb{I}$$

$$\Rightarrow n \text{ must be multiple of } 24$$

4. **Ans (A,C,D)**

Let equation of circle is

$$\lambda(x \cos \alpha_1 + y \sin \alpha_1 - p_1)(x \cos \alpha_2 + y \sin \alpha_2 - p_2) +$$

$$\mu(x \cos \alpha_2 + y \sin \alpha_2 - p_2)(x \cos \alpha_3 + y \sin \alpha_3 - p_3) +$$

$$\gamma(x \cos \alpha_3 + y \sin \alpha_3 - p_3)(x \cos \alpha_1 + y \sin \alpha_1 - p_1) = 0$$

.... (1)

Apply coefficient of $x^2 =$ coefficient of y^2 and

coefficient of $xy = 0$

$$\Rightarrow \lambda(\cos \alpha_1 \cos \alpha_2 - \sin \alpha_1 \sin \alpha_2) + \mu(\cos \alpha_2 \cos \alpha_3 -$$

$$\sin \alpha_2 \sin \alpha_3)$$

$$+ \gamma(\cos \alpha_3 \cos \alpha_1 - \sin \alpha_3 \sin \alpha_1) = 0$$

$$\text{and } \lambda(\sin \alpha_1 \cos \alpha_2 + \cos \alpha_1 \sin \alpha_2) + \mu(\sin \alpha_2 \cos \alpha_3 +$$

$$\cos \alpha_2 \sin \alpha_3)$$

$$+ \gamma(\sin \alpha_3 \cos \alpha_1 + \cos \alpha_3 \sin \alpha_1) = 0$$

$$\Rightarrow \lambda \cos(\alpha_1 + \alpha_2) + \mu \cos(\alpha_2 + \alpha_3) + \gamma \cos(\alpha_3 + \alpha_1) = 0$$

.... (2)

$$\text{and } \lambda \sin(\alpha_1 + \alpha_2) + \mu \sin(\alpha_2 + \alpha_3) + \gamma \sin(\alpha_3 + \alpha_1) = 0$$

.... (3)

from (1), (2) and (3) we get equation circle is

$$\begin{vmatrix} L_1 L_2 & L_2 L_3 & L_3 L_1 \\ \cos(\alpha_1 + \alpha_2) & \cos(\alpha_2 + \alpha_3) & \cos(\alpha_3 + \alpha_1) \\ \sin(\alpha_1 + \alpha_2) & \sin(\alpha_2 + \alpha_3) & \sin(\alpha_3 + \alpha_1) \end{vmatrix} = 0$$

Because it passes through origin so

$$\begin{vmatrix} p_1 p_2 & p_2 p_3 & p_3 p_1 \\ \cos(\alpha_1 + \alpha_2) & \cos(\alpha_2 + \alpha_3) & \cos(\alpha_3 + \alpha_1) \\ \sin(\alpha_1 + \alpha_2) & \sin(\alpha_2 + \alpha_3) & \sin(\alpha_3 + \alpha_1) \end{vmatrix} = 0$$

$$\Rightarrow p_1 p_2 \sin(\alpha_1 - \alpha_2) + p_2 p_3 \sin(\alpha_2 - \alpha_3) + p_3 p_1 \sin(\alpha_3 - \alpha_1) = 0$$

$$\begin{vmatrix} p_1 p_2 & p_2 p_3 & p_3 p_1 \\ \cos \alpha_3 & \cos \alpha_1 & \cos \alpha_2 \\ \sin \alpha_3 & \sin \alpha_1 & \sin \alpha_2 \end{vmatrix} = 0$$

5. **Ans (A,B,D)**

$$\text{Use } -1 \leq \sin(\alpha + \beta) \leq 1$$

$$-1 \leq \sin(\alpha - \beta) \leq 1$$

$$\begin{aligned} \text{Now } \frac{1}{4} - (\sin\alpha + \cos\beta)^2 - \cos^2\alpha \sin^2\beta &= \frac{1}{4} - \\ (\sin^2\alpha + \cos^2\beta + 2\sin\alpha \cos\beta) - \cos^2\alpha \sin^2\beta &= \\ = \frac{1}{4} + 1 - \sin^2\alpha - \cos^2\beta - \cos^2\alpha \sin^2\beta &= \\ = \frac{5}{4} - \sin^2\alpha - \cos^2\beta - (1 - \sin^2\alpha)(1 - \cos^2\beta) &= 0 \end{aligned}$$

6. **Ans (B,C)**

$$g'(x) = 2f(2x) - f(x)$$

$$\text{because } g'(1) = 0 \text{ and } g'(2) = 0$$

$$\Rightarrow f(2) = 0 \text{ and } f(4) = 0 \{ \because f(1) = 0 \}$$

$$\Rightarrow f(x) = (x-1)(x-2)(x-4)$$

$$g'(x) = 15x^3 - 49x^2 + 42x - 8$$

SECTION-I(ii)

7. **Ans (A)**

$$(P) \quad P(E_1) = \frac{1}{7}$$

$$(Q) \quad P(E_2) = \frac{6}{7}$$

$$(R) \quad P(E_3) = \frac{3}{7}$$

$$(S) \quad P(E_4) = \frac{4}{7}$$

8. **Ans (B)**

$$(P) \quad \frac{-b}{3a} \text{ is root of } f(x) = 0$$

$$(Q) \quad \left(\frac{-d}{a} \right)^{1/3} \text{ is root of } f(x) = 0$$

$$(R) \quad \left(\frac{-c}{3d} \right) \text{ is root of } f(x) = 0$$

$$(S) \quad \text{Take } f(x) = a(x-m)^3 \text{ and compare coefficient}$$

9. **Ans (C)**

$$(P) \quad a^2 + b^2 + c^2 = ab + bc + ca \Rightarrow (a-b)^2 + (b-c)^2 + (c-a)^2 = 0 \Rightarrow a = b = c \Rightarrow \Delta ABC \text{ is equilateral triangle}$$

$$(Q) \quad a^2 + 2b^2 + c^2 = 2bc + 2ab \Rightarrow (a-b)^2 + (b-c)^2 = 0 \Rightarrow a = b = c \Rightarrow \Delta ABC \text{ is equilateral triangle}$$

$$(R) \quad a^2 + b^2 + c^2 = \sqrt{2}a(b+c) \Rightarrow 2a^2 + 2b^2 + 2c^2 - 2\sqrt{2}a(b+c) = 0$$

$$\Rightarrow (a^2 - 2\sqrt{2}ab + 2b^2) + (a^2 - 2\sqrt{2}ac + 2c^2) = 0$$

$$\Rightarrow (a - \sqrt{2}b)^2 + (a - \sqrt{2}c)^2 = 0$$

$$a = \sqrt{2}b = \sqrt{2}c$$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\Rightarrow A = 90^\circ \text{ \& } B = C$$

$$B = C = 45^\circ$$

$$(S) \quad a^2 + b^2 + c^2 = ca + ab\sqrt{3} \Rightarrow 4a^2 + 4b^2 + 4c^2 - 4ac - 4ab\sqrt{3} = 0$$

$$\Rightarrow (3a^2 - 4\sqrt{3}ab + 4b^2) + (a^2 - 4ac + 4c^2) = 0$$

$$\Rightarrow (\sqrt{3}a - 2b)^2 + (a - 2c)^2 = 0$$

$$a = \frac{2}{\sqrt{3}}b = 2c \Rightarrow b = \frac{\sqrt{3}}{2}a \text{ and } c = \frac{a}{2}$$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} = 0$$

$$\Rightarrow A = 90^\circ \text{ \& } C = 30^\circ$$

$$B = 60^\circ$$

10. **Ans (D)**

Here 24 matrices are possible value of determinants can be $-8, -4, -2, 2, 4, 8$

(A) Possible non-negative values of $|A|$ are 2, 4, 8 $\Rightarrow$ sum = 14

(B) Sum of these 24 determinants is 0

(C) Mod. $(\det(A))$ is least $|A| = \pm 2 \Rightarrow$

$$|\text{adj}(\text{adj}(\text{adj}A))| = |A|^{(n-1)^3} = \pm 2$$

(D) Least value of $\det(A)$ is -8 Now $|4A^{-1}| =$

$$16 \frac{1}{|A|} = \frac{16}{-8} = -2$$

SECTION-II

1. Ans (0.00)

Let $(2 \sec \theta, \tan \theta)$ be the point on hyperbola $\frac{x^2}{4} - \frac{y^2}{1} = 1$, at which the tangent is being drawn

$$\text{Equation of tangent : } \frac{x \sec \theta}{2} - y \tan \theta = 1 \quad \dots$$

(1)

let (h, k) be the mid-point of chord PQ of the given ellipse

$$\text{Equation of chord with given mid-point: } T = S_1$$

$$\Rightarrow \frac{hx}{4} + \frac{ky}{1} = \frac{h^2}{4} + \frac{k^2}{1} \quad \dots(2)$$

Comparing (1) and (2)

$$\Rightarrow \frac{\left(\frac{\sec \theta}{2}\right)}{\left(\frac{h}{4}\right)} = \frac{-\tan \theta}{k} = \frac{1}{\left(\frac{h^2}{4} + k^2\right)}$$

$$\Rightarrow \sec \theta = \frac{2h}{h^2 + 4k^2}$$

$$\text{and } \tan \theta = \frac{-4k}{h^2 + 4k^2}$$

$$\Rightarrow \frac{4h^2}{(h^2 + 4k^2)^2} - \frac{16k^2}{(h^2 + 4k^2)^2} = 1$$

$$\Rightarrow 4h^2 - 16k^2 = (h^2 + 4k^2)^2$$

$$\text{locus of mid-point of PQ: } (x^2 + 4y^2)^2 = 4(x^2 - 4y^2)$$

$$\text{Point of intersection with x-axis } \Rightarrow y = 0$$

$$\Rightarrow x^4 = 4x^2$$

$$\Rightarrow x = 0, 2, -2$$

But $(0, 0)$, $(2, 0)$ and $(-2, 0)$ can not be the mid point of any chord

2. Ans (7.00)

Differentiate functional equation w.r.t y we get

$$f' \left(\frac{x}{y} \right) \left(\frac{-x}{y^2} \right) = f(x) \left(\frac{-1}{y^2} \right) - x \left(\frac{y^2 f'(y) - 2yf(y)}{y^4} \right) + \frac{x}{y^2} \quad \dots(1)$$

put $y = 1$ in equation (1) we get

$$-x f'(x) = -f(x) - x (f'(1) - 2f(1)) + x \quad \dots(2)$$

Put $x = y$ in the given functional equation we get $f(1) = -1$

Now put value of $f(1) = -1$ in (2) we get

$$-x f'(x) = -f(x) - x \Rightarrow x \frac{dy}{dx} - y = x$$

$$\Rightarrow \frac{dy}{dx} - \frac{y}{x} = 1 \Rightarrow \frac{y}{x} = \int \frac{dx}{x} \Rightarrow \frac{y}{x} = \log_e$$

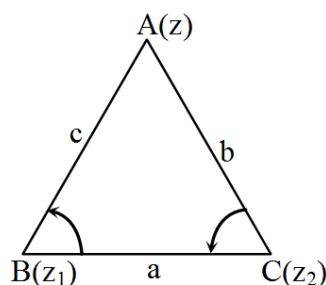
$$x + c \quad \dots(3)$$

$$\text{put } x = 1 \text{ in equation (3) we get } \frac{-1}{1} = 0 + c$$

$$\Rightarrow c = -1$$

$$f(x) = x \ln x - x$$

3. Ans (5.00)



$$\frac{\tan \frac{\theta_1}{2}}{\tan \frac{\theta_2}{2}} = \frac{3}{2}$$

$$\Rightarrow \frac{\sqrt{\frac{(s-a)(s-c)}{s(s-b)}}}{\sqrt{\frac{(s-a)(s-b)}{s(s-c)}}} = \frac{3}{2} \Rightarrow \frac{s-c}{s-b} = \frac{3}{2} \Rightarrow 2s -$$

$$2c = 3s - 3b$$

$$\Rightarrow 6b = 2s + 4c \Rightarrow b - c = \frac{a}{5} \Rightarrow AC - AB = \frac{a}{5}$$

$$\Rightarrow 2ae = BC = a$$

$$2a = \frac{BC}{5} = \frac{a}{5} \Rightarrow e = 5$$

4. Ans (173.00)

$$D_6 - \frac{1}{5} D_6 - \frac{1}{5} D_6 + (4! - 3! - 3! + 2!) = 265 - 53 - 53 + (14) = 173$$

5. Ans (3.00)

$$px^3 + qx^2 + qx + p = 0$$

$$\Rightarrow p(x+1)(x^2 + 1 - x) + qx(x+1) = 0$$

$$\Rightarrow (x+1)(px^2 + (q-p)x + p) = 0$$

$$\text{Let } g(x) = px^2 + (q-p)x + p$$

$$\Rightarrow g(2) = 3p + 2q < 0 \text{ and } g(3) = 7p + 3q > 0$$

$\Rightarrow$ One root of $g(x) = 0$ lies in $(2, 3)$ and other

root lies in $\left(\frac{1}{3}, \frac{1}{2}\right)$ because product of root equals to one

$$\Rightarrow \alpha = -1, \beta \in \left(\frac{1}{3}, \frac{1}{2}\right) \text{ and } \gamma \in (2, 3)$$

$$\Rightarrow [|\alpha|] + [|\beta|] + [|\gamma|] = 1 + 0 + 2 = 3$$

6. Ans (0.25)

Put $x = 1$ we get $5^n = a_0 + a_1 + a_2 + a_3 + a_4 + a_5 + \dots$

$$\text{put } x = e^{i\frac{2\pi}{5}} \text{ we get } 0 = a_0 + a_1 e^{i\frac{2\pi}{5}} + a_2 e^{i\frac{4\pi}{5}} + \dots$$

$$\text{put } x = e^{i\frac{4\pi}{5}} \dots$$

$$\text{put } x = e^{i\frac{6\pi}{5}} \dots$$

$$\text{put } x = e^{i\frac{8\pi}{5}} \text{ we get } 0 = a_0 + a_1 e^{i\frac{8\pi}{5}} + a_2 e^{i\frac{16\pi}{5}} + \dots$$

After simplification we get

$$a_0 + a_5 + a_{10} + \dots = a_1 + a_6 + a_{11} + \dots = a_2 + a_7 +$$

$$a_{12} + \dots = a_3 + a_8 + a_{13} + \dots =$$

$$a_4 + a_9 + a_{14} + \dots = \frac{5^n}{5} = 5^{n-1}$$

$$\Rightarrow \text{Required equals to } \frac{5^{n-1}}{4 \times 5^{n-1}} = 0.25$$

7. Ans (7.00)

$$\text{Let } g(x) = (\ln x) (\ln x) \dots \infty$$

$$g(x) =$$

$$\begin{cases} 0 & 1 < x < e \\ 1 & x = e \\ \infty & e < x \leq 3 \end{cases}$$

$$f(x) =$$

$$\begin{cases} x & 1 < x < e \\ \frac{x}{2} & x = e \\ 0 & e < x \leq 3 \end{cases}$$

Hence $f(x)$ is non differentiable at $x = e$.

8. Ans (8.00)

$$x - 4 = 2 \cos \theta \Rightarrow x = 2 \cos \theta + 4$$

$$y = 3 \sin \theta \Rightarrow y = 3 \sin \theta$$

$$\therefore z = \frac{x^2}{4} + \frac{y^2}{9} = \frac{(2 \cos \theta + 4)^2}{4} + \sin^2 \theta$$

$$z = \frac{4 \cos^2 \theta + 4 \sin^2 \theta + 16 \cos \theta + 16}{4}$$

$$= \frac{20 + 16 \cos \theta}{4} = 5 + 4 \cos \theta$$

$$Z_{\max} - Z_{\min} = (9 - 1) = 8$$

Alternatively :

$$\frac{x^2}{4} + \frac{y^2}{9} + 4 - 2x = 1$$

$$\frac{x^2}{4} + \frac{y^2}{9} = 2x - 3$$

$$\Rightarrow z = 2x - 3 \text{ where } x \in [2, 6]$$

$$Z_{\max} - Z_{\min} = 9 - 1 = 8$$